

SMART ANTENNAE FOR Wi-Fi NETWORKS

By using smart antennae, it's possible to provide greater range, higher reliability and predictable speeds for Wi-Fi users

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Wi-Fi networks have evolved rapidly over the last decade to become a preferred means of local wireless networking today. With people carrying more and more Wi-Fi-enabled mobile devices such as laptops, netbooks and smart-phones, these networks are now being used extensively for a variety of purposes in homes, offices, colleges, retail shops and other premises.

With such extensive Wi-Fi usage scenarios which involve wireless Internet access, operating mission-critical and real-time voice/video applications wirelessly, etc, expectations of reliability, predictability, high throughput and increased range are high from Wi-Fi networks.

Reliability in Wi-Fi networks generally refers to the availability of relatively strong Wi-Fi signal at any place supposed to be covered by the network. Predictability, on the other hand, refers to the consistency of throughput as required by real-time applications running on Wi-Fi clients. Range refers to the distance up to which the desired Wi-Fi signal is available.

However, a Wi-Fi signal is subjected to attenuation with increase in distance or in presence of obstacles. Also, it may get corrupted by intermixing with an interfering signal. Signal attenuation and corruption, along with the use of conventional single omnidirectional antenna access point (Fig. 1) that spreads/receives energy in/ from all directions (Fig. 2), constitute the main reasons for low reliability, predictability and range of the Wi-Fi signal.



Fig. 1: A conventional access point with omnidirectional antenna



Fig. 2: An omnidirectional access point blasts energy in all directions as well as receives energy from all directions

Although the latest 802.11n Wi-Fi standard provides faster raw speeds and some improvements in reliability, predictability and range aspects by using more than one omnidirectional antenna (upper limit is four by the standard), Wi-Fi infrastructure vendors are coming up with smart antennae and beamforming systems using smart antennae to provide greater range, higher reliability and predictable speeds for users/applications.

Smart antennae were initially used

for radar systems and military communication systems, but advances in digital signal processing and antenna technology have made it possible to use them in Wi-Fi and cellular networks (such as 3G and WiMAX).

How does a smart antenna system work?

Basically, a smart antenna system consists of an array of antenna elements. These elements can be arranged linearly, circularly or in some other fashion and are controlled by an on-chip processor or a separate software controller. Further, the array elements can be directional or omnidirectional in nature.

By controlling the antenna elements of the array, the smart antenna system can form various optimised antenna radiation patterns for transmission and reception of signals over the air. A particular optimised radiation pattern in which the signal is transmitted or received from a particular direction is generally called 'beamforming.' Optimised radiation patterns including the beamforming (Fig. 3) aim at:

1. Delivering strong signal strength at the receiver. In line-of-sight (LOS) environments, the strong signal can be ensured by the beamforming radiation pattern directed toward the receiver. In non-LOS cases, the optimised radiation pattern takes care of multipath propagation characteristics for the signal and is generally different from a directional beamforming pattern.

2. Receiving the strong strength of the transmitted signal by aligning to proper directions in the radiation pattern. Also, the radiation pattern ensures that it does not receive in-